

Philosophes. Version en utilisant l'état des voisins.

EXTENDS *Naturals*

CONSTANT *N*

Philos $\triangleq 0 \dots N - 1$

gauche(*i*) $\triangleq (i + 1) \% N$ philosophe à gauche du philo n°*i*
droite(*i*) $\triangleq (i + N - 1) \% N$ philosophe à droite du philo n°*i*

Hungry $\triangleq \text{"H"}$

Thinking $\triangleq \text{"T"}$

Eating $\triangleq \text{"E"}$

VARIABLES

etat $i \rightarrow \text{Hungry}, \text{Thinking}, \text{Eating}$

TypeOK $\triangleq \Box (etat \in [\text{Philos} \rightarrow \{\text{Hungry}, \text{Thinking}, \text{Eating}\}])$

exclusion

ExclMutuelle $\triangleq \Box (\forall i \in \text{Philos} : (etat[i] \triangleq \text{Eating}) \Rightarrow (etat[\text{gauche}(i)] \neq \text{Eating} \wedge etat[\text{droite}(i)] \neq \text{Eating}))$

vivacité individuelle

VivaciteIndividuelle $\triangleq \forall i \in \text{Philos} : etat[i] \triangleq \text{Hungry} \leadsto etat[i] \triangleq \text{Eating}$

vivacité globale

VivaciteGlobale $\triangleq (\exists i \in \text{Philos} : etat[i] \triangleq \text{Hungry}) \leadsto (\exists j \in \text{Philos} : etat[j] \triangleq \text{Eating})$

Init $\triangleq$

$\wedge etat = [i \in \text{Philos} \mapsto \text{Thinking}]$

demander(*i*) $\triangleq$

$\wedge etat[i] = \text{Thinking}$

$\wedge etat' = [etat \text{ EXCEPT } ![i] = \text{Hungry}]$

manger(*i*) $\triangleq$

$\wedge etat[i] = \text{Hungry}$

$\wedge etat' = [etat \text{ EXCEPT } ![i] = \text{Eating}]$

$\wedge etat[\text{gauche}(i)] \neq \text{Eating}$

$\wedge etat[\text{droite}(i)] \neq \text{Eating}$

penser(*i*) $\triangleq$

$\wedge etat[i] = \text{Eating}$

$$\wedge \textit{etat}' = [\textit{etat} \text{ EXCEPT } ![i] = \textit{Thinking}]$$

$$\begin{aligned} \textit{Next} &\triangleq \\ &\exists i \in \textit{Philos} : \vee \textit{demander}(i) \\ &\quad \vee \textit{manger}(i) \\ &\quad \vee \textit{penser}(i) \end{aligned}$$

$$\textit{Fairness} \triangleq \text{TRUE } \text{A changer}$$

$$\begin{aligned} \textit{Spec} &\triangleq \\ &\wedge \textit{Init} \\ &\wedge \Box[\textit{Next}]_{\langle \textit{etat} \rangle} \\ &\wedge \textit{Fairness} \end{aligned}$$
